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Lecture 23

Linearization of Renormalization Operator & stability analysis of RG flow

Here we analyze the stability of RG flow around fixed points where $\hat{R}\{\vec{K}^*\} \longrightarrow \{\vec{K}^*\}$

i.e. $K_1^* = \hat{R}_1\{K_1^*, K_2^*, \dots\}$

$$K_2^* = \hat{R}_2\{K_1^*, K_2^*, \dots\}$$

so on.

In the vicinity of K^* , the R.G. eqⁿ for $\Delta\vec{K} = \vec{K} - \vec{K}^*$, is given by the Jacobian matrix $\hat{\mathcal{L}}_b = \left[\frac{\partial \hat{R}_i}{\partial K_j} \right]_{\text{at } \vec{K}^*}$ as

$$\Delta K_j' = \sum_i \left[\frac{\partial \hat{R}_i}{\partial K_j} \right]_{\vec{K}^*} \Delta K_i$$

$$\Rightarrow \Delta\vec{K}' = \hat{\mathcal{L}}_b(\hat{R})|_{\vec{K}^*} \Delta\vec{K}$$

[expanding only upto 1st order of ΔK_i]

In matrix notation it is,

$$\begin{bmatrix} \Delta K'_1 \\ \Delta K'_2 \\ \Delta K'_3 \\ \vdots \end{bmatrix} = \underbrace{\begin{bmatrix} \left. \frac{\partial \hat{R}_1}{\partial K_1} \right|_{\vec{K}^*} & \left. \frac{\partial \hat{R}_1}{\partial K_2} \right|_{\vec{K}^*} & \left. \frac{\partial \hat{R}_1}{\partial K_3} \right|_{\vec{K}^*} & \dots \\ \left. \frac{\partial \hat{R}_2}{\partial K_1} \right|_{\vec{K}^*} & \left. \frac{\partial \hat{R}_2}{\partial K_2} \right|_{\vec{K}^*} & \left. \frac{\partial \hat{R}_2}{\partial K_3} \right|_{\vec{K}^*} & \dots \\ \left. \frac{\partial \hat{R}_3}{\partial K_1} \right|_{\vec{K}^*} & \left. \frac{\partial \hat{R}_3}{\partial K_2} \right|_{\vec{K}^*} & \left. \frac{\partial \hat{R}_3}{\partial K_3} \right|_{\vec{K}^*} & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}}_{\hat{\mathcal{L}}_b} \begin{bmatrix} \Delta K_1 \\ \Delta K_2 \\ \Delta K_3 \\ \vdots \end{bmatrix}$$

We need to diagonalize $\hat{\mathcal{L}}_b$ next.
 Let $\{|q_m^{(R)}\rangle\}$ be the set of right eigen vectors & $\{|q_m^{(L)}\rangle\}$ be the set of left eigen vectors of $\hat{\mathcal{L}}_b$, then

$$\Rightarrow \hat{\mathcal{L}}_b |q_m^{(R)}\rangle = \Lambda_m^{(b)} |q_m^{(R)}\rangle$$

$$\& \langle q_m | \hat{\mathcal{L}}_b = \langle q_m | \Lambda_m^{(b)}$$

where $\langle q_m^{(L)} | = (|q_m^{(L)}\rangle)^T$ & $\Lambda_m^{(b)}$ are eigen values of $\hat{\mathcal{L}}_b$.

If $\hat{L}_b$ is diagonalizable, then the matrix of all right eigen vectors

$$S = [|q_1^{(R)}\rangle \ |q_2^{(R)}\rangle \ |q_3^{(R)}\rangle \ \dots]$$

is the similarity transformation matrix which diagonalizes $\hat{L}_b$

$$\Rightarrow \hat{S}^{-1} \hat{L}_b \hat{S} = \text{Diag}(\{\Lambda_m^{(b)}\})$$

The left & right eigen vectors will always form a biorthogonal set where $\langle q_m^{(L)} | q_n^{(R)} \rangle = \delta_{mn}$

$$\Rightarrow \hat{L}_b = \sum_m \Lambda_m^{(b)} |q_m^{(R)}\rangle \langle q_m^{(L)}|$$

$\{|q_m^{(R)}\rangle\}$ will form a basis for all column vectors as they are linearly independent eigen kets.

Any column vector $|v\rangle$ in the eigen basis of $\hat{L}_b$ is $|\tilde{v}\rangle = \hat{S}^{-1}|v\rangle$

In the basis of $\{|q_m^{(R)}\rangle\}$ we can write

$$\Delta \vec{K} = \sum_m h_m^{(0)} |q_m^{(R)}\rangle$$

where $h_m^{(0)}$ are coefficients of linear expansion of $\Delta \vec{K}$ in $\hat{\mathcal{L}}_b$ basis.

Thus we can think of ' $h_m^{(0)}$ ' as the coordinates of $\Delta \vec{K}$ in some general basis spanned by the vectors $\{|q_m\rangle\}$ (need not be orthogonal)

$$\Rightarrow \hat{\mathcal{L}}_b \Delta \vec{K} = \sum_m h_m^{(0)} \Lambda_m^{(b)} |q_m^{(R)}\rangle$$

$$\& \hat{\mathcal{L}}_b^l \Delta \vec{K} = \sum_m h_m^{(0)} (\Lambda_m^{(b)})^l |q_m^{(R)}\rangle$$

If we had initially started with the block spins of length b^l , then acting once with $\hat{\mathcal{L}}_b^l$ would be equivalent to acting $\hat{\mathcal{L}}_b$ 'l' times (upto first order of ΔK_i).

$$\Rightarrow \hat{\mathcal{L}}_b^l \equiv \hat{\mathcal{L}}_b^e \quad \& \quad (\Lambda_m^{(b)})^l \equiv \Lambda_m^{(b^l)}$$

$$\Rightarrow \hat{\mathcal{L}}_b^l \Delta \vec{K} = \Delta \vec{K}^{(l)} = \sum_m h_m^{(0)} (\Lambda_m^{(b)})^l |q_m^{(R)}\rangle$$

$$\Rightarrow \Delta \vec{K}^{(l)} = \sum_m h_m^{(l)} |q_m^{(R)}\rangle$$

$$\text{where } h_m^{(l)} = h_m^{(0)} (\Lambda_m^{(b)})^l$$

$$\Rightarrow \underline{h_m^{(l+1)} = \Lambda_m^{(b)} h_m^{(l)}}$$

Thus under action of RG transformation near the fixed point, ' h_m ' will just undergo scaling by a factor of $\Lambda_m^{(b)}$. Some h_m will increase if $\Lambda_m^{(b)} > 1$ and decrease if $\Lambda_m^{(b)} < 1$.

$$\text{Now } \Delta \vec{K}^{(l)} = \sum_m h_m^{(l)} |q_m^{(R)}\rangle$$

$$\Rightarrow \underline{h_m^{(l)} = \langle q_m^{(L)} | \Delta \vec{K}^{(l)} \rangle}$$

$\{h_m^{(l)}\}$ are usually called scaling fields. They are very important as they do not mix under the action of RG transformation near the fixed point.

Since $(\Lambda_m^{(b)})^l = \Lambda_m^{(b^l)}$, we define the scaling exponent ' λ_m ' using,

$$\underline{\Lambda_m^{(b)} = b^{\lambda_m}}$$

$$\text{Thus } (\Lambda_m^{(b)})^l = \Lambda_m^{(b^l)} \Rightarrow (b^{\lambda_m})^l = (b^l)^{\lambda_m}$$

$$\Rightarrow \underline{h_m^{(l)} = h_m^{(0)} (b^l)^{\lambda_m}}$$

$$\& \underline{\Delta \vec{K}^{(l)} = \sum_m h_m^{(0)} (b^l)^{\lambda_m} |q_m^{(R)}\rangle}$$

$$\& \underline{\hat{\mathcal{L}}_b |q_m^{(R)}\rangle = b^{\lambda_m} |q_m^{(R)}\rangle}$$

$$\& \underline{\langle q_m | \hat{\mathcal{L}}_b = \langle q_m | b^{\lambda_m}}$$

Scaling form for Correlation length

The correlation length is a property of the system defined by the coupling constants $(\vec{K})$. But in the vicinity of fixed point, the main governing parameters of the system are the scaling fields $\{h_m\}$.

$$\text{So } \xi(\{K^{(l)}\}) = \frac{\xi(\{\vec{K}\})}{b^l} \text{ in the vicinity}$$

of a fixed point, becomes,

$$\xi(\{h_m^{(l)}\}) = \frac{\xi(\{h_m\})}{b^l}$$

$$\Rightarrow \xi(\{h_m\}) = b^l \xi(h_1^{(0)} b^{l\lambda_1}, h_2^{(0)} b^{l\lambda_2}, h_3^{(0)} b^{l\lambda_3}, \dots)$$

choose 'l' such that $h_1^{(0)} b^{l\lambda_1} = 1$

$$\Rightarrow b^l = (h_1^{(0)})^{-1/\lambda_1}$$

Thus we get,

$$\begin{aligned}\mathcal{E}_g(\{h_m\}) &= (h_1^{(0)})^{-1/\lambda_1} \mathcal{E}_g(1, h_2^{(0)} (h_1^{(0)})^{-\frac{\lambda_2}{\lambda_1}}, h_3^{(0)} (h_1^{(0)})^{-\frac{\lambda_3}{\lambda_1}}, \dots) \\ \Rightarrow \mathcal{E}_g(\{h_m\}) &= (h_1^{(0)})^{-1/\lambda_1} \chi \left[\frac{h_2^{(0)}}{(h_1^{(0)})^{\frac{\lambda_2}{\lambda_1}}}, \frac{h_3^{(0)}}{(h_1^{(0)})^{\frac{\lambda_3}{\lambda_1}}}, \frac{h_4^{(0)}}{(h_1^{(0)})^{\frac{\lambda_4}{\lambda_1}}}, \dots \right]\end{aligned}$$

This is the scaling form for ' $\mathcal{E}_g$ '.

$$\text{Let } \phi_m = \frac{\lambda_m}{\lambda_1}$$

$$\Rightarrow \mathcal{E}_g(\{h_m\}) = (h_1^{(0)})^{-1/\lambda_1} \chi \left[\frac{h_2^{(0)}}{h_1^{\phi_2}}, \frac{h_3^{(0)}}{h_1^{\phi_3}}, \frac{h_4^{(0)}}{h_1^{\phi_4}}, \dots \right]$$

Thus under RG operation,

$$h_m^{(0)} \longrightarrow h_m^{(l)} = \frac{h_m^{(0)}}{(h_1^{(0)})^{\phi_m}}$$